

WILL-RAG - Animish Final Contribution

Venkata Naga Sai Animish Kocharlakota | Branch: animish-evaluation

1. Work Completed

- Set up and tested the WILL-RAG pipeline in the required Conda/Python environment.
- Added and tested an additional research paper: “The Attention System of the Human Brain: 20 Years After” (P05).
- Ran PDF extraction, text cleaning, chunking, DPR passage embeddings, FAISS indexing, retrieval, and RAG answer generation.
- Added the P05 metadata entry to data/metadata/papers.csv and rebuilt the index after chunking.
- Built the FAISS index successfully with 56 vectors.
- Created and pushed the GitHub branch animish-evaluation with the metadata and evaluation documentation.
- Updated the team Trello board with completed and remaining evaluation tasks.

2. Retrieval Evaluation

Test question: What are the main attention networks described in the paper?

Check	Result
Retrieved chunks relevant?	Generally yes; all Top-5 results came from the correct paper.
Paper metadata correct?	Yes for the tested retrieval.
Page references present?	Yes.
DOI correct?	Yes: 10.1146/annurev-neuro-062111-150525.
Ranking quality	Relevant overall, but Rank 2 was more directly useful than Rank 1.

3. RAG Answer Evaluation

- Final answer generation completed successfully after a working project API key was provided.
- The generated answer was grounded in retrieved sources and cited the correct paper, pages, and DOI.
- The answer was clear and understandable for a non-expert user.
- The answer was mostly correct, but it focused on the later dorsal/ventral attention and cingulo-opercular/frontoparietal networks and did not clearly mention the original alerting, orienting, and executive framework. This makes it grounded but not fully comprehensive.

4. Insufficient-Evidence Test

Test question: Does this paper provide evidence that drinking coffee cures Alzheimer’s disease?

- Passed: the system correctly stated that the paper contains no evidence about coffee, caffeine, or Alzheimer’s disease.
- The model did not hallucinate a treatment claim and explicitly limited its answer to the available evidence.

- The answer was clear for a non-expert user.
- Observation: the retriever still returned five chunks even though the query was unrelated. A relevance threshold or abstention mechanism could improve retrieval behaviour.

5. Issues Found

- The first chunking attempt produced no usable chunks because the new paper was not yet registered in papers.csv.
- Running build_index.py with an empty chunks.jsonl produced “No chunks found in chunks.jsonl.”
- The .env file initially had the wrong format and had to be corrected to OPENAI_API_KEY=<key>.
- Initial RAG generation attempts were blocked by OpenAI API credit/authentication issues; this was later resolved with a working project key.
- Retrieval ranking may need improvement because the most directly relevant chunk was not always ranked first.
- For unrelated questions, Top-5 chunks are still returned even when there is no strong evidence match.

6. Current Status

Completed: ingestion, metadata, chunking, DPR embeddings, FAISS indexing, retrieval testing, final RAG generation, answer grounding/source checks, non-expert clarity check, insufficient-evidence test, Trello update, documentation, Git commit and push.

Remaining: additional questions and/or team-provided dataset testing if required by the project lead, plus any final report refinements requested by the team.

7. Project Links

Trello Board:

<https://trello.com/invite/b/6a857c4afe82ed74c1a5ba72/ATTI892d3281f98fb88bce3d40b1b028d893A3BCB7C9/my-trello-board>

GitHub Repository: <https://github.com/s4078157/WILL-Project-RAG>

Branch: animish-evaluation

8. Team Update Summary

Suggested message to the team:

“I have completed and pushed my current evaluation work to the animish-evaluation branch. I tested the full pipeline with my own research paper through extraction, cleaning, chunking, DPR embeddings, FAISS indexing, retrieval, and RAG generation. I checked retrieval relevance, paper/page/DOI metadata, answer grounding and clarity, and I also ran an insufficient-evidence test, which passed. I documented the issues I found and updated Trello. The main observations are that retrieval ranking can still be improved and unrelated queries still return Top-5 chunks, so a relevance threshold may be useful. My current evaluation report is available on my branch.”